

London Metropolitan University

School of Computing and Digital Media

CS7P01NM: MSc Project

A FRIENDLY GUIDE TO COMPLETE THE MSC PROJECT

Compiled by

Dr Kaneeka Vidanage

NEXT Campus

Contents

1.	Introduction.....	4
1.1	Why you are required to complete the project	5
1.2	Benefits to be gained by students	6
2.	Learning aims and goals	6
3.	Preparation and allocation of the project	7
3.1	How to select a project	8
3.2	Type of projects.....	9
3.2.1	Software Development Projects	9
3.2.2	Research Projects	9
3.2.3	Artificial Intelligence and Machine Learning Projects	9
3.2.4	Data Science Projects.....	10
3.2.5	Cyber Security Projects.....	10
4.	Carrying out your project.....	11
4.1	How should students plan for their tasks and deliverables	13
4.1.1	General Project Timeline	14
4.1.2	AI-Specific Project Timeline	14
4.2	Meeting your supervisor.....	15
4.2.1	The Student Perspective.....	16
4.3	NEXT Moodle.....	16
5.	MSc Project Report.....	17
5.1	Report Format	17
5.2	Developing the Project Proposal	17
5.2.1	Problem Statement.....	17
5.2.2	Rationale	18
5.2.3	Aim	18
5.2.4	Objectives	19
5.2.5	Research Questions	19
5.2.6	Expected Deliverables	20
5.3	Project Planning and Management.....	20
5.3.1	Project Schedule.....	20
5.3.2	Gantt Chart.....	21
5.3.3	Risk Assessment	21
5.4.	Literature Review	22
5.4.1	Purpose of the Literature Review	22
5.4.2	Sources of Literature	22

5.4.3 Critical Analysis.....	23
5.4.4 Research Gap	23
5.5. Methodology	23
5.5.1 Research Methodology	23
5.5.2 Proposed Methodology	24
5.5.3 Benefits of the Proposed Methodology.....	24
5.5.4 Ethical Considerations	25
5.6. Analysis and Design.....	25
5.6.1 Requirements Analysis	25
5.6.2 Design Activities.....	25
5.7 Implementation	26
5.7.1 Development Environment	26
5.7.2 Solution Development	26
5.7.3 Challenges Encountered.....	26
5.8 Testing and Validation	27
5.8.1 Testing Strategy	27
5.8.2 Testing Types.....	27
5.8.3 Test Evidence.....	27
5.8.4 Validation.....	28
5.9 Evaluation and Performance Analysis.....	28
5.9.1 Evaluation Framework.....	28
5.9.2 Performance Metrics	29
5.9.3 Benchmark Comparison.....	29
5.9.4 Discussion	30
5.9.5 Evaluation Against Objectives.....	30
5.9.6 Limitations	30
5.10 Conclusion and Future Work	31
5.10.1 Conclusion	31
5.10.2 Contributions	31
5.10.3 Future Work	31
5.11 References	32
5.12 Final Dissertation Structure.....	32
5.14 Report Writing Style	33
5.13 Viva Presentation	34
6. BCS project requirements.....	35
7. Plagiarism / Turnitin	36

8. Reassessment	37
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1. Introduction

The MSc Project is a significant part (60 credits) of your master's degree programme, and it will give you the opportunity to apply the knowledge acquired during the taught elements of the course to current research and commercial problems.

The aim of this guide is to give you a general overview of the project and provide useful information on how to carry out the project. You should aim to complete the project on time, and to the highest of professional standards. Contained alongside this module guide is a module specification, an Ethics form, the marking scheme and the grading criteria.

The MSc project is an exercise that may take a variety of forms depending on the nature of the project. It should provide students with the experience of planning and bringing to fruition a major piece of individual work. It aims to encourage and reward individual inventiveness and application of effort. Although the project is not expected to be a wholly original piece of research, the research element should be of a good contribution and scientific relevance and sufficiently challenging that it cannot be dismissed as mere implementation of a standard solution of a common computing problem (for example, a database implementation for a common application area). The scope of the project is not only to complete a well-defined piece of work in a professional manner, but also to place the work into the context of the current state of the art within the subject discipline.

The project may take a variety of forms and it will involve the specification, design, implementation, validation and documentation of a substantial piece of individual application that will fall in areas such as software engineering principles, database development, intelligent system, cyber and network security, mobile computing, data mining, big data, cloud computing, IoT and distributed systems. Projects that involve literature surveys, or empirical field studies, or evaluation of one or more techniques, methods or tools are also acceptable. However, projects that are wholly or mainly literature surveys, or comparative studies based on case studies with no practical outcomes are discouraged. Also you will need to make sure to:

- Select a subject that you wish to investigate
- It is vital that you choose a **subject that inspires you**, motivation is essential.
- This is going to be **your own independent piece** of work so you must be driven from within. Nobody is going to be constantly on your back telling you what to do next.
- The bottom line is that **you** are in charge.

Also, there is a balance to be struck between: going for too broad a subject in which you lose direction, and ultimately present a muddled and superficial dissertation, and too narrow a subject where you are unable to find others who have written on it before.

This looks like a much bigger task than any you have had to deal with before. The key words are:

- Knowledge
- Planning
- Preparation

1.1 Why you are required to complete the project

There are some serious objectives here. In most of your academic work you are required to learn **specific facts** or understand given ideas. Then you are assessed on your **knowledge** and **understanding** by identifying the facts and applying the ideas in coursework and examinations. So far there has been little opportunity for you to decide for yourself what is it that **you want to discover** and develop **your own fresh ideas**. Also, projects give students an opportunity to apply the knowledge acquired during the taught elements of the course to current research problems. They also help to develop important project management and communication skills that are highly likely valued by employers and international research bodies. You will be asked to devise, manage and produce to a deadline a substantial piece of work. In doing so, you will have to analyse and evaluate a wide range of appropriate and relevant data and most importantly express your own ideas on the matter under investigation. Finally, you will need to communicate the outcomes of your investigation effectively to your supervisors and peers by means of the written report and presentation (viva). It should be the most satisfying piece of work in your degree.

1.2 Benefits to be gained by students

Some tangible benefits can be gained from completing the project successfully:

- A deep understanding of new subject of your own choice
- Application of fundamental concepts within a business environment
- Producing artefacts with scientific relevance and commercial significance
- Interaction and engagement with business partners
- Opportunities of finding good jobs
- A good application for a PhD programme to carry on the subject.

2. Learning aims and goals

The module aims to encourage and reward individual inventiveness and application of effort. It also aims to allow students to have an opportunity to assimilate the knowledge they gained in their courses and extend this knowledge to new area of application; to apply newly acquired knowledge and techniques to a specific problem using established research techniques and methods; to determine the framework on the project according to a set of specifications relevant to the subject study; to manage an extended piece of work by confining the problem within the constraints of time and available resources; to research effectively the background material on the topic using a variety of sources and to develop ability to conduct critical analysis and draw conclusions; to be able to develop a framework/model of scientific relevance and commercial significance which is crucial for future career and employability; to develop the ability to produce detailed specifications and design frameworks relevant to the problem of investigation in the subject related to the industry; to demonstrate the originality in the application of new knowledge and skills; to report your findings and communicate the work to others by means of verbal and appropriate documentation techniques; to raise awareness potential business development opportunities in an area pertinent to the topic.

3. Preparation and allocation of the project

An MSc project is a substantial and extensive investigation of a challenging topic within the subject area of the MSc programme. It is mainly assessed through a major written project report, which explains the full scope of the project, including its aims and objectives, background study, requirements analysis, research methodology, design or experimental work, implementation, testing, evaluation, conclusions, and recommendations. Therefore, students are supervised by qualified academics with relevant subject knowledge while they carry out their MSc projects.

Preparation for the MSc project normally begins during Semester 3, when students attend Research Methods workshops delivered by the project module leader or relevant academic staff. These workshops help students understand the expectations of the MSc project and develop important skills such as critical thinking, research planning, project topic selection, academic writing, proposal development, and ethical awareness. During these workshops, students are encouraged to start identifying suitable project ideas and developing their initial project proposal.

After completing the Research Methods workshops, students are required to select their supervisors based on their proposed project area and supervisor availability. Immediately after supervisor selection, students must submit their project proposal along with the required Ethics Form. The proposal should normally include the project title, aims and objectives, rationale, background reading, proposed methodology, expected outcomes, and references. The supervisor will then provide guidance to further develop and refine the project work after the proposal submission.

Students are also required to complete and submit the Ethics Form, which will be provided to them. The proposal and Ethics Form must be submitted as instructed through NEXT Moodle. Students should note that they cannot begin their project work until the proposal has been reviewed and the Ethics approval has been granted. The module leader/supervisor will coordinate the proposal review, supervisor allocation, and ethics approval process to ensure that students can begin their MSc projects in a proper and timely manner.

3.1 How to select a project

Think about what kind of project you want to do. Think about what kind of projects you have enjoyed in the past and what may motivate you the most during your MSc studies. This is going to be your own independent piece of work so you must be driven from within. Nobody is going to be constantly on your back telling you what to do next.

While preparing for the MSc project you need to:

- Make sure it is going to be a feasible project
- Find out whether you are going to be able to get the necessary technical information or opinions of others that you need
- Identify the required resources and make sure the software, hardware and other equipment are available to carry out your project. You should consult with the module leader on the availability of the required resources.

Students might have ideas:

- Discuss them with your tutors and Readers, Professors and business partners within research centres.
- Read articles and professional journals
- Search in the internet and library database
- Access past MSc reports on NEXT Moodle

This will help you to narrow the subject further. Remember, there is always a balance to be struck between going for too broad a subject in which you lose direction, and ultimately present a muddled and superficial dissertation; and too narrow a subject where you are unable to find others who have written on it before.

3.2 Type of projects

Projects may take several forms depending on the programme pathway and research interests.

3.2.1 Software Development Projects

Projects associated with the process of designing, developing, testing, and evaluating software that is designed to solve a certain problem.

Examples:

- Hospital management software solutions
- Mobile applications
- Enterprise software solutions
- Cloud-based platforms

3.2.2 Research Projects

Projects concerned with the investigation of a certain problem based on experimentation, comparisons, or empirical research.

Examples:

- Comparison between machine learning techniques' performance
- Vulnerability assessment for cybersecurity
- Research on human-computer interaction

3.2.3 Artificial Intelligence and Machine Learning Projects

Projects associated with:

- Predictive modeling
- Natural language processing
- Computer vision
- Deep learning
- Generative AI

- Recommender systems

3.2.4 Data Science Projects

Projects associated with:

- Data analysis
- Forecasting
- Business intelligence
- Predictive analytics

3.2.5 Cyber Security Projects

Projects associated with:

- Threat detection
- Intrusion prevention
- Cybersecurity assessments
- Digital forensics

Regardless of the chosen topic, you should show originality and critical analysis of the issue.

4. Carrying out your project

Before the commencement of the final semester, students are required to attend a series of project preparation workshops conducted by the module leader. These workshops are designed to guide students in understanding the MSc project requirements, identifying suitable project areas, developing research ideas, understanding proposal expectations, and preparing for the supervision process.

At the end of the workshop series, students will be given the opportunity to select their preferred supervisor based on their proposed project area and supervisor availability. Following the supervisor selection process, students are required to submit their project proposal for review and approval.

Once the final semester starts, students are expected to:

- Conduct intensive reading on materials relevant to their selected topic, identify gaps, and formulate these gaps as key research elements.
- Develop a clear understanding of the problem domain and background of the selected project area.
- Attend the scheduled workshops or project-related sessions to discuss common issues, share ideas, and receive further guidance where applicable.
- Attend the scheduled supervision sessions and use them effectively to discuss progress, challenges, methodology, results, and report development.
- Make an appointment to meet the supervisor as early as possible to agree on the project plan, frequency of meetings, milestones, deliverables, and other important matters.

Students are strongly advised to attend and make proper use of all supervision sessions, as lack of communication with the supervisor may affect the progress and quality of the project. The first meeting with the supervisor is expected to help students clarify the project specification, confirm the proposed direction, discuss any areas of uncertainty, and agree on the supervision plan.

Students are required to develop a sound project management plan and identify appropriate milestones using standard project management techniques such as a Gantt chart, critical path

analysis, or other suitable planning tools. A clear and organised project plan will be used as a benchmark by the supervisor to assess the student's progress throughout the module.

Since the MSc project is a major piece of work carried out over a period of time, students must keep all project-related information well organised. This may include literature search records, notes from background reading, datasets, model designs, software or hardware designs, experimental results, analysis outputs, simulations, supervisor meeting notes, and draft documentation.

Keeping information properly organised will help both the student and the supervisor to monitor progress. It will also support the final writing stage of the dissertation, as students will be able to refer back to their work, decisions, findings, and thought processes more easily. Poor organisation may slow down progress and affect the quality and flow of the final report.

Students must be aware of all project requirements, tasks, deliverables, and phases in order to plan and complete the project successfully. Therefore, students are expected to:

- Develop a project from an initial or unstructured idea through a detailed analysis of the problem area.
- Apply relevant tools, techniques, theories, and practical knowledge gained from taught modules.
- Demonstrate systematic understanding of knowledge and critical awareness of current problems, new developments, and/or new insights in the selected specialist area.
- Justify the selected methodology, tools, techniques, models, datasets, and overall approach.
- Demonstrate how the project contributes to knowledge or practice. This contribution may be achieved through algorithmic improvements, novel applications, framework development, experimental evaluation, explainability, optimisation, or the generation of new insights.
- Show a strong understanding of the project context and explain how the selected approach addresses the identified problem.
- Understand the role of deliverables at different stages of the project cycle.

- Reflect on the relationship between design decisions and the satisfaction of user, system, or research requirements.
- Demonstrate a clear interpretation of results for a wide range of audiences, especially where real-life data is limited or unavailable.
- Test and evaluate the selected solution to prove its conformance to the project specification.
- Critically evaluate the project outcomes, limitations, and possible areas for future improvement.

For MSc Artificial Intelligence projects, students should place particular emphasis on experimentation, model evaluation, benchmark comparison, reproducibility, and critical analysis. Where applicable, students should compare their results with existing models, techniques, datasets, benchmarks, or baselines and justify the significance of their findings.

Students are expected to take responsibility for the progress of their project by attending supervision sessions, maintaining regular communication with the supervisor, using Moodle and WhatsApp communication channels appropriately, meeting agreed milestones, keeping accurate records of project activities, and submitting the required deliverables within the given deadlines.

4.1 How should students plan for their tasks and deliverables

After understanding the project specification, students should carefully plan how they will complete the project and achieve the intended objectives. Before starting the main project work, students are expected to prepare a clear project plan with suitable milestones and deliverables. The timeline may vary depending on the nature, scope, and complexity of the project. However, the following general timeline may be used as a guide.

4.1.1 General Project Timeline

Weeks Suggested Activity

Week	Task
1 - 3	Understand the problem area, conduct background reading, and develop domain understanding.
4 - 6	Select the most suitable methodology or solution approach. Identify and analyse requirements using a structured or unstructured approach.
7 - 9	Design the proposed model, framework, system, ontology, or algorithm.
10 - 11	Implement the proposed solution and carry out testing of the main functionalities.
12 - 13	Finalise the dissertation writing, review the report, and prepare for submission.

4.1.2 AI-Specific Project Timeline

Students can follow an AI-specific project pathway, as these projects normally involve dataset preparation, model development, experimentation, evaluation, and reporting.

Weeks Suggested AI Project Activity

Week	Task
1 - 3	Conduct the literature review, understand the problem domain, and identify suitable datasets.
4 - 5	Collect, clean, preprocess, and prepare the dataset for modelling.
6 - 8	Develop the baseline model and initial AI/ML solution.

9 - 10	Conduct model experimentation, hyperparameter tuning, optimisation, and benchmark comparison.
11	Evaluate the model using suitable performance metrics and, where relevant, carry out explainability analysis.
12 - 13	Finalise the dissertation writing, interpret results, discuss limitations, and prepare for submission.

Students are strongly advised to maintain regular notes throughout the project, including literature findings, design decisions, experimental results, supervisor feedback, and changes made during the project. Students should also consult their supervisor regularly and make effective use of the scheduled supervision sessions to ensure that progress is maintained according to the agreed project plan.

4.2 Meeting your supervisor

Formal supervision will commence from the beginning of the final semester. During the supervision period, students will be provided with six supervision sessions with their allocated supervisor. These sessions are intended to support students in discussing their project progress, clarifying academic or technical concerns, receiving feedback, reviewing milestones, and improving the quality of their project work. Supervisors will also provide a common introduction to each dissertation chapter, explaining the expected structure, purpose, and key areas to be covered. This will help students understand how to organise and develop their final report in a clear and academically appropriate manner.

In addition to the scheduled supervision sessions, students may communicate with their supervisor through the Moodle chat option for project-related queries and clarifications. However, students are expected to use these communication channels professionally and responsibly.

Supervisors are asked to record attendance at every meeting.

4.2.1 The Student Perspective

What you want from your supervisor is support and guidance. You need to agree the frequency and format of meetings and having done so, stick to it; nothing annoys busy professionals more than preparing and making themselves available for meeting to find that the student fails to turn up without notice or explanation. The meeting is the confirmation of your engagement with your supervisor, and the time should be used as efficiently as possible. You will need to make sure you are prepared before each meeting. Make sure you know what goals were set at the previous session and be ready to discuss your progress against your project plan and take the opportunity to ask all the questions. The bottom line is to keep your supervisor on your side and make sure you inform him/her of your progress, ask for advice, ask if your progress is satisfactory and discuss your future plans. Your supervisor will almost certainly make comments that are uncomfortable to hear. Accept criticism gracefully, it is for your own good and you will deliver a better report for it. You are more likely to get higher marks if you follow your supervisor guidance.

4.3 NEXT Moodle

Copies of all documentation relating to the module plus useful information are available via NEXT Moodle. All announcements made by the module leader will automatically reach you via Moodle. Also, the MSc report submission will be made through NEXT Moodle and there is no need for students to submit hard copies.

5. MSc Project Report

5.1 Report Format

As a general rule, the body of the dissertation should be between 12,000-15,000 words this will normally corresponds to about 40-60 pages if you include some diagrams. We don't have prescriptive style/format, but students should choose a font that is easy to read (normally 10 or 12) and are encouraged to use one and a half line spacing. You should include appendices for additional material not central to the report (e.g. questionnaires, screenshots) and these will be in addition to 12,000-15,000 pages for the main body.

5.2 Developing the Project Proposal

The project proposal can be described as the backbone of the MSc Project. Proper planning of the project proposal increases the likelihood of project success.

5.2.1 Problem Statement

The problem statement highlights the problem that the project aims at solving.

A good problem statement ought to do the following:

- Describe the problem clearly.
- Explain the stakeholders involved.
- State why the problem is relevant and significant.
- List limitations.

Example

Phishing attacks cost organizations a lot financially and reputation-wise. The current methods of detection of these kinds of attacks fail in many cases because they cannot detect sophisticated attacks hence low detection rate.

What Examiners Want to See

- Clear description of the problem.
- Relevance of the problem statement.
- Significance of the problem statement.

5.2.2 Rationale

The rationale explains why the project needs to be done.

Students are supposed to:

- State the importance of the problem.
- Explain why the current approaches are inadequate.
- Justify the need for further investigation.
- Show how the project could add value to the existing knowledge.

The students need to prove the value of the project.

Common Mistake:

Starting the topic rather than giving reasons for doing the project.

5.2.3 Aim

The aim refers to the overall goal of the project.

The project should have one main aim.

Example

- Develop an intelligent phishing-detection framework using machine learning.

5.2.4 Objectives

The objectives refer to the various activities that will be undertaken to achieve the aim.

Objectives are supposed to be:

- Specific
- Measurable
- Achievable
- Relevant
- Time-bounded

Example

- Carry out a comprehensive literature review.
- Find suitable data sets.
- Build baseline machine learning models.
- Design and implement the suggested system.
- Measure and test the efficiency of the system.
- Propose areas for improvement.

5.2.5 Research Questions

Projects aimed at solving research problems usually have some research questions.

Examples

- What is the most efficient machine learning model for detecting phishing?
- What are the weaknesses and strengths of the proposed model compared to other models?

5.2.6 Expected Deliverables

Students ought to state their expected deliverables.

Expected deliverables may include the following:

- **Project Proposal** – including the motivation, aim, objectives, and Gantt chart.
- **Literature Review** – critical review of relevant research and existing solutions.
- **Project Planning and Management** – including methodology, project schedule, risk & ethical assessment, and resource management.
- **Project Design, Implementation, Analysis and Evaluation** – development of the proposed solution, results, critical analysis, and performance evaluation.
- **Conclusion and Future Work** – summary of achievements, limitations, and recommendations for further development.
- **References** – formatted in accordance with the required referencing style.

5.3 Project Planning and Management

Successful projects require effective planning and monitoring.

Poor planning is one of the most common reasons for project failure.

5.3.1 Project Schedule

Students should create a project schedule at the beginning of the project.

The schedule should identify:

- Activities
- Deliverables
- Milestones
- Deadlines

5.3.2 Gantt Chart

A Gantt Chart should be included within the dissertation and maintained throughout the project.

Typical activities include:

- Literature Review
- Requirements Analysis
- Design
- Development
- Testing
- Evaluation
- Dissertation Writing

The Gantt Chart demonstrates project management skills and allows supervisors to monitor progress.

5.3.3 Risk Assessment

Risk assessment identifies factors that may affect project success.

Students should maintain a risk register throughout the project.

Example

Risk	Probability	Impact	Mitigation
Dataset unavailable	Medium	High	Use alternative datasets
Hardware failure	Low	High	Maintain backups
Time constraints	Medium	High	Weekly monitoring
API restrictions	Medium	Medium	Use local implementation

What Examiners Look For

- Awareness of risks
- Realistic mitigation strategies
- Evidence of project management

5.4. Literature Review

The literature review establishes the academic foundation of the project.

This chapter demonstrates that students understand the current state of knowledge and can critically evaluate existing research.

5.4.1 Purpose of the Literature Review

The literature review should answer:

- What has already been done?
- What methods have been used?
- What limitations exist?
- What research gaps remain?

5.4.2 Sources of Literature

Students should primarily use:

- IEEE Xplore
- ACM Digital Library
- Springer
- Elsevier
- Scopus
- Web of Science

Sources should be current, relevant, and credible.

5.4.3 Critical Analysis

A literature review should not be a collection of summaries.

Students should:

- Compare studies.
- Analyse strengths and weaknesses.
- Identify trends.
- Evaluate methodologies.

Example Comparison Table

Study	Method	Strength	Limitation
Author A	CNN	High Accuracy	Small Dataset
Author B	Transformer	Robust Performance	High Cost
Author C	Random Forest	Fast Training	Lower Accuracy

5.4.4 Research Gap

The chapter should conclude by identifying a gap that motivates the proposed project.

This is one of the most important sections of the dissertation.

5.5. Methodology

The methodology explains how the project will be conducted.

This chapter should justify every major decision.

5.5.1 Research Methodology

Possible methodologies include:

- Experimental Research

- Design Science Research
- Agile Development
- Case Study Research
- Comparative Evaluation

Students must justify why the selected methodology is appropriate.

5.5.2 Proposed Methodology

Students should explain:

- Overall workflow
- Tools and technologies
- Datasets
- Algorithms
- Frameworks
- Evaluation methods

Flow diagrams are strongly encouraged.

5.5.3 Benefits of the Proposed Methodology

Students should explain why their chosen approach is preferable.

Benefits may include:

- Higher accuracy
- Better scalability
- Reduced complexity
- Faster execution
- Better maintainability
- Improved explainability

A comparison with alternative approaches is recommended.

5.5.4 Ethical Considerations

Students should consider:

- Data privacy
- Security
- Fairness
- Bias
- Transparency
- Responsible use of AI

Ethical considerations are particularly important for AI and data-driven projects.

5.6. Analysis and Design

This chapter transforms project requirements into a solution design.

5.6.1 Requirements Analysis

Students should identify:

Functional Requirements

Services and functions provided by the system.

Non-Functional Requirements

Performance, reliability, usability, security, scalability, and maintainability requirements.

5.6.2 Design Activities

Depending on the project, students may include:

- System Architecture
- UML Diagrams
- Database Design
- Network Design
- AI Architecture

- Framework Design

Design decisions should be justified.

5.7 Implementation

This chapter explains how the solution was developed.

5.7.1 Development Environment

Describe:

- Programming languages
- Frameworks
- Libraries
- Hardware
- Development tools

5.7.2 Solution Development

Students should explain:

- Major system components
- Algorithms implemented
- Software modules
- AI models developed

Screenshots, diagrams, and architecture illustrations should be included where appropriate.

5.7.3 Challenges Encountered

Students should discuss:

- Technical challenges
- Resource limitations
- Development difficulties

and explain how these challenges were addressed.

5.8 Testing and Validation

Testing demonstrates that the solution works correctly.

Validation demonstrates that the solution solves the intended problem.

5.8.1 Testing Strategy

Students should explain their overall testing approach.

5.8.2 Testing Types

- **Unit Testing**
Testing individual components.
- **Integration Testing**
Testing interactions between components.
- **System Testing**
Testing the complete system.
- **User Acceptance Testing**
Testing with stakeholders or end users.

5.8.3 Test Evidence

Students should provide:

- Test plans
- Test cases
- Results
- Screenshots
- Logs

Example

Test Case	Expected Result	Actual Result	Status
Login	User authenticated	User authenticated	Pass

5.8.4 Validation

Validation should demonstrate that project objectives have been achieved.

For AI projects, validation may include:

- Cross-validation
- Benchmark testing
- Performance comparisons

5.9 Evaluation and Performance Analysis

Many students confuse testing with evaluation.

Testing proves that the system functions correctly.

Evaluation proves that the solution is effective and valuable.

5.9.1 Evaluation Framework

It is not enough to achieve something in doing your MSc project by way of software development or by conducting some experiments. You also need to demonstrate the worth of what you have achieved by some kind of independent standard other than your own satisfaction with what you have done. With a software development you can do this by conducting an evaluation with the help of some third parties. Evaluation is different from testing your software. The aim of testing is to verify that your software does what is designed to do. The aim of the evaluation is to validate that your software fulfils the project's requirements. A minimum evaluation might be a checklist comparison of what the original requirements were and what/how you succeeded in implementing them.

Students should explain:

- Evaluation criteria
- Performance indicators
- Success measures

5.9.2 Performance Metrics

Software Engineering Projects

- Reliability
- Maintainability
- Response Time
- Usability

AI and Machine Learning Projects

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Cyber Security Projects

- Detection Rate
- False Positive Rate
- Response Time

5.9.3 Benchmark Comparison

Students should compare results against:

- Existing solutions
- Published research
- Baseline models
- Industry standards

5.9.4 Discussion

Students should critically discuss:

- Strengths
- Weaknesses
- Unexpected findings
- Practical implications

5.9.5 Evaluation Against Objectives

Each objective should be revisited.

Objective	Achieved	Evidence
Objective 1	Yes	Literature Review
Objective 2	Yes	Developed Framework
Objective 3	Yes	Evaluation Results

5.9.6 Limitations

Every project has limitations.

Students should openly discuss:

- Dataset limitations
- Time constraints
- Computational limitations
- Scope restrictions

Honest reflection is viewed positively by examiners.

5.10 Conclusion and Future Work

The conclusion should clearly demonstrate what has been achieved.

5.10.1 Conclusion

Students should summarize:

- The problem addressed
- The methodology adopted
- The solution developed
- Key findings

5.10.2 Contributions

Students should identify:

- **Academic Contributions**
Knowledge generated.
- **Technical Contributions**
Systems, models, frameworks, or algorithms developed.
- **Industrial Contributions**
Practical benefits and applicability.

5.10.3 Future Work

Students should propose realistic future improvements.

Examples include:

- Larger datasets
- Additional features
- Improved models
- Real-world deployment
- Scalability enhancements

5.11 References

Your report may cite a wide range of sources (e.g. papers or web sites that you have used) as background and context of the work. Sources are cited at the relevant point in the text and full source information is given in the references section. There are a variety of acceptable citation and referencing style, but the most commonly used styles are the Harvard style and the IEEE style which are recommended in the MSc projects.

5.12 Final Dissertation Structure

The recommended dissertation structure is:

1. Abstract
2. Acknowledgements
3. Table of Contents
4. Chapter 1 – Introduction
5. Chapter 2 – Literature Review
6. Chapter 3 – Methodology
7. Chapter 4 – Analysis and Design
8. Chapter 5 – Implementation
9. Chapter 6 – Testing and Validation
10. Chapter 7 – Evaluation and Performance Analysis
11. Chapter 8 – Conclusion, Limitations and Future Work
12. References
13. Appendices

Chapter 1 should normally start with a short introduction to the problem you are addressing and your aims and objectives, give a short review of the context and describe what follows in the main body of the report.

Chapter 2 should normally include a critical review of the pertinent literature review so the reader would understand what you are building on. You may also describe techniques, guidelines and even existing products if relevant to what you will be presenting later. It is important that this review is written in your words throughout, reads as a coherent and connected piece of writing, shows the relevance of the material presented to the problem being addressed and provide some critique / analysis of the material and its applicability to the problem. In essence it is your analysis and understanding that we are interested in, how you build on existing work, understand its limitations and gaps, select from available methods/tools with some justifications, and present that coherently.

It is important you select your references carefully in your review. It is not sufficient to find 15 web sites which seem to have something relevant to say. Sources should be authoritative, accurate and preferably should still be around in five years' time. Academic papers and books usually meet these criteria but also some web sources are acceptable.

This structure ensures that all MSc projects consistently demonstrate rationale, objectives, risk assessment, deliverables, literature review, methodology justification, testing, evaluation, critical analysis, limitations, and future work while meeting postgraduate-level expectations.

5.14 Report Writing Style

The main point to make is to present materials clearly and concisely and in as an objective fashion as possible. Your personal impressions and feelings should rarely come onto it. You should normally avoid using expressions like "I did this" and instead report the work in a passive voice (it was done). However, where you are genuinely voicing an individual opinion, you may use the first person. Also, while the passive voice is normal for scientific writing it is not used universally, so don't feel forced into a style that you may find awkward. The main thing is clarity and objectivity.

5.13 Viva Presentation

Each MSc student will be given a date to present his/her work to the supervisor and a second marker. This usually happens within one weeks from the submission date. The duration of the viva is 20 minutes; students will be given 10 minutes to present their work, and the remainder will be used for questions and feedback. To provide some guidance, the presentation should contain the following items:

Students should be prepared to discuss:

- Problem Statement
- Rationale
- Literature Review
- Methodology
- Design Decisions
- Implementation
- Testing
- Evaluation
- Limitations
- Future Work

For AI projects, students should additionally be prepared to justify:

- Dataset selection
- Model choice
- Hyperparameter settings
- Evaluation metrics
- Bias and fairness considerations
- Explainability methods

This AI-specific guidance is mainly relevant to AI/ML experimental projects, data science projects, and AI engineering projects. However, MSc AI students may follow different project types depending on their selected topic. Therefore, the viva questions and presentation focus will depend on the nature, aim, methodology, and outcomes of the individual project.

6. BCS project requirements

The British Computer Society defines the particular requirements that projects on Computing programmes should meet. These are given below for your information:

General project requirements: within their curricular projects, students are expected to deliver a good quality project. as outlined above, prior to starting the project and while they are doing their Research and development skills' module, students are being prepared for the project and their awareness towards the project is raised through a good guidance in terms of projects' expectation, type and format of project, selection of projects through engaging with tutors and they are run through the assessment criteria mentioned above.

As described above the report should include:

- Aims and objectives, rationale and elucidation of the problem
- Evidence of a thorough relevant literature review with criticality
- Full description of research hypothesis through software or framework architecture.
- Full justification and choice of appropriate methodology and approach
- Discussion of results and critical evaluation and appraisal of own work; explaining the rationale behind design and full implementation of functionalities.
- An update to date and relevant references

Postgraduate project requirements: Project at the postgraduate level should be more advanced compared to Undergraduate project; project should contain an in depth investigation of a problem leading to innovative ideas appropriate to a master's degree.

Postgraduate projects must give students the opportunity to demonstrate;

- A good knowledge and awareness of current problems and extend this knowledge to new area of application
- A comprehensive understanding of techniques; apply newly acquired knowledge and techniques to a specific problem using established research techniques and methods.

- An ability to produce detailed specifications and design frameworks relevant to the problem investigation in the subject related to the industry with originality in the application of new knowledge and skills; good discussion of how results are produced with a detailed interpretation of the acquired knowledge.
- The subject should be challenging and complex and requires creativity in the field of research
- Good skills to effectively communicate the work to others including specialists and non-specialist in the field by means of verbal and appropriate documentation techniques.
- Demonstrate self-awareness, planning, problem solving, and originality in the application of new knowledge and professional skills.
- Critical self-evaluation of the whole process.

7. Plagiarism / Turnitin

You are reminded that all work submitted as part of the requirements must be expressed in your own words and incorporate your own ideas and judgements. Plagiarism, which is the presentation of other person's thoughts or words as though they were your own – this must be avoided, with particular care taken in the report written in your own time. Direct quotations from the published or unpublished work of others must always be clearly identified as such being placed inside quotation marks, and a full reference to their source must be provided in the proper form. Remember that a series of short quotations from several such sources, if not clearly identified as such, constitutes plagiarism just as much as a single unacknowledged long quotation from a single source. Equally, if you summarise another person's ideas or judgements you must refer to that person in your text, and include the work referred to in your bibliography. Similarly, if you copy an illustration from any publication (whether or not you modify it), you must always acknowledge the source by means of a statement. Prior to submitting your report, you will need to make sure you run it through Turnitin which is available on WebLearn. If the outcomes are above 8% then it is of a concern and therefore students need to consult with their supervisors and seek advice from the module leader.

8. Reassessment

If you fail the module in the main sit then you will have an opportunity for a re-sit in the following semester. However, there is no formal supervision during the re-sit. Generally speaking, a failure in the project process will require you to undertake further development work on your analysis and solution. Please note that in the re-sit your mark will be capped to 50%.